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Phytoremediation as a Sustainable Tool for Pollution Control: Thematic Advances, Chronological Development, Challenges, and Future Perspectives

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Abstract

Phytoremediation is a plant-based approach for the removal, immobilization, transformation, or attenuation of contaminants in soil, sediment, and water. Its principal modes include phytoextraction, phytostabilization, rhizofiltration, phytodegradation, phytovolatilization, and rhizodegradation. The field has progressed from early demonstrations of plant-assisted contaminant removal to integrated approaches involving plant–microbe interactions, soil amendments, omics technologies, genetic improvement, and hybrid remediation systems. This review synthesizes the thematic and chronological development of phytoremediation, with particular emphasis on the mechanistic basis of contaminant uptake and detoxification, plant selection, environmental controls, wastewater applications, and emerging enhancement strategies. A critical comparison with selected conventional remediation technologies shows that phytoremediation can offer low site disturbance and potentially lower costs, but its effectiveness is constrained by contaminant bioavailability, rooting depth, plant growth rate, climate, site heterogeneity, and the need for safe management of contaminated biomass. Importantly, phytoremediation should not be regarded as universally superior to physicochemical remediation; rather, it is most appropriate where contaminant concentrations, site conditions, remediation endpoints, and timescales are compatible with plant-based treatment. Recent research is increasingly focused on microbe-assisted remediation, amendments, nanotechnology, genetic and multi-omics approaches, modelling, and biomass valorization. Future progress will depend on field-scale validation, standardized performance indicators, life-cycle assessment, regulatory clarity, and integrated remediation designs that combine phytoremediation with complementary technologies.

Keywords: Phytoremediation, Heavy Metals, Organic Pollutants, Plant–Microbe Interactions, Contaminated Biomass, Sustainable Remediation.

1. Introduction

Environmental contamination by heavy metals, petroleum hydrocarbons, pesticides, industrial chemicals, and other persistent pollutants is a major constraint on ecosystem function and human well-being. Conventional remediation technologies can provide rapid and predictable contaminant removal, but many physicochemical methods require substantial energy, infrastructure, chemical inputs, excavation, or off-site disposal. Phytoremediation provides an in situ or semi-in situ alternative in which plants and their associated rhizosphere microorganisms contribute to contaminant removal, immobilization, transformation, or attenuation. The approach is particularly attractive for large, moderately contaminated areas where ecological restoration and contaminant management can be pursued simultaneously¹⁻⁴.



The performance of phytoremediation depends on the interaction among contaminant properties, soil or water chemistry, plant physiology, root architecture, microbial activity, and climatic conditions. For heavy metals, the objective is generally extraction or stabilization because metals cannot be biologically degraded. For many organic contaminants, plant enzymes and rhizosphere microorganisms can contribute to transformation or degradation. This distinction is fundamental when evaluating remediation claims and selecting appropriate treatment endpoints.

The present review reorganizes the submitted manuscript into a critical synthesis of the development, mechanisms, applications, limitations, and emerging technologies of phytoremediation. Particular attention is given to the gap between laboratory performance and field-scale implementation, an issue repeatedly identified in recent reviews⁵⁻⁹.

2. Scope and Literature Approach

This article is structured as a narrative critical review rather than a systematic review or meta-analysis. The revised synthesis emphasizes peer-reviewed literature on phytoremediation mechanisms, heavy-metal and organic pollutant remediation, wastewater treatment, plant-microbe interactions, molecular and omics approaches, amendments, genetic improvement, modelling, and post-harvest biomass management. The original manuscript's broad

chronological and thematic framework was retained, while unsupported generalizations, non-peer-reviewed sources, and repetitive statements were reduced. Quantitative publication counts are not presented as bibliometric results because the original submission did not provide a reproducible database search strategy.

3. Thematic Advances in Phytoremediation

3.1. Major Phytoremediation Modes

Phytoremediation is an umbrella term encompassing several processes that differ in their treatment objectives (Table 1). Phytoextraction involves uptake of contaminants, particularly metals, followed by translocation to harvestable plant tissues. Phytostabilization limits contaminant mobility and bioavailability through root-associated processes, sorption, precipitation, or sequestration. Rhizofiltration uses roots and associated microbial communities to remove contaminants from water. Phytodegradation or phytotransformation refers mainly to enzymatic transformation of organic contaminants within plant tissues. Rhizodegradation is driven largely by microorganisms stimulated by plant roots. Phytovolatilization transfers selected contaminants or their transformation products to the atmosphere and therefore should be assessed as a contaminant-transfer process rather than automatically as detoxification^{1, 5, 6}.

Table 1. Major modes of phytoremediation and their principal mechanisms, targets, and limitations

Mode	Primary process	Typical targets	Principal limitation
Phytoextraction	Root uptake and translocation to harvestable tissues	Cd, Zn, Pb, Ni and other metals	Slow removal; biomass management
Phytostabilization	Immobilization and reduction of mobility/bioavailability	Metals and selected organics	Contaminants remain on site
Rhizofiltration	Root-associated sorption and uptake from water	Metals and dissolved contaminants	Requires sustained root biomass and water management
Phytodegradation / phytotransformation	Plant-mediated transformation	Organic pollutants	Dependent on contaminant chemistry and plant metabolism
Rhizodegradation	Microbial degradation stimulated by roots	Petroleum hydrocarbons and other organics	Strongly site- and microbiome-dependent
Phytovolatilization	Uptake and release of volatile forms or transformation products	Selected elements and organics	Potential transfer to atmosphere

3.2. Mechanistic Basis

Plant-assisted remediation begins with contaminant availability at the root–soil or root–water interface. Root exudates, rhizosphere pH, organic acids, redox reactions, and microbial metabolites can modify contaminant speciation and availability. For metals, transport across the root is followed by intracellular chelation, sequestration, vacuolar compartmentalization, or translocation through the vascular system. Transporter families including heavy-metal ATPases, ZIP proteins, NRAMP proteins, and ATP-binding cassette transporters participate in metal homeostasis and detoxification. Phytochelatins, metallothioneins, cell-wall binding, and antioxidant systems contribute to tolerance^{5, 6, 10}.

These mechanisms also explain why high total contaminant concentrations do not necessarily translate into high plant uptake. Metal speciation, pH, cation competition, organic matter, mineralogy, moisture, and redox status can substantially alter the fraction available for uptake. Consequently, remediation performance should be evaluated using both contaminant concentration and bioavailability, together with plant biomass and contaminant accumulation.

3.3. Plant Selection and Rhizosphere Processes

Plant selection is a central determinant of phytoremediation success. Hyperaccumulators can achieve high tissue concentrations but often have low biomass and relatively slow growth. Conversely, high-biomass grasses, shrubs, and woody species may provide greater total contaminant removal or stabilization despite lower tissue concentrations. Selection should therefore consider biomass productivity, root architecture, contaminant tolerance, accumulation factor, translocation factor, climatic adaptation, water requirements, harvesting feasibility, and ecological compatibility^{5, 7, 11}.

The rhizosphere provides an important interface between plants and contaminant-degrading microorganisms. Plant roots supply carbon substrates and signalling compounds that can stimulate microbial populations, while beneficial bacteria and fungi may improve plant growth, nutrient acquisition, stress tolerance, and

contaminant transformation. Microbe-assisted phytoremediation is consequently emerging as an important strategy for overcoming limitations of plant growth under contaminated conditions^{6, 9}.

3.4. Molecular and Multi-Omics Approaches

Genomics, transcriptomics, proteomics, metabolomics, and integrated multi-omics approaches have improved understanding of plant responses to contaminants. These tools can identify transporters, regulatory networks, antioxidant pathways, chelation systems, and metabolic signatures associated with tolerance or accumulation. Their greatest value for phytoremediation lies in connecting molecular responses with measurable traits such as uptake, translocation, biomass production, and stress tolerance. However, omics-derived candidates require physiological and field validation before they can be translated into practical phytoremediation technologies.

4. Applications

4.1. Soil and Sediment Remediation

Soil remediation remains the major application of phytoremediation, particularly for metal-contaminated agricultural land, mine-impacted soils, and industrial sites. Phytoextraction is appropriate where contaminants are sufficiently bioavailable and plants can accumulate them without severe phytotoxicity. Phytostabilization is often more realistic for highly contaminated sites where complete removal is impractical. Recent work also supports combined plant–biochar and other hybrid approaches to improve contaminant immobilization, plant establishment, and overall treatment performance^{7, 12, 13}.

4.2. Wastewater and Aquatic Systems

Aquatic plants and constructed wetland systems can support phytoremediation through rhizofiltration, nutrient uptake, sediment retention, and interactions with microbial biofilms. Such systems are particularly relevant for polishing municipal, agricultural, and industrial wastewater. Their performance depends on hydraulic loading, contaminant concentration, plant growth, seasonal conditions, and harvesting frequency. Recent reviews emphasize the potential of native aquatic species for heavy-metal removal, while also highlighting the need for careful biomass management¹⁴.

5. Comparative Assessment with Conventional Remediation

Phytoremediation should be selected on the basis of site-specific objectives rather than a general assumption that it is cheaper or more sustainable than all conventional technologies. Excavation and soil washing can provide rapid contaminant removal but may generate secondary waste or require substantial energy and infrastructure. Thermal and electrokinetic methods can be

effective for specific contaminants and soil types but may have high energy or operational requirements. Bioremediation and phytoremediation are generally less disruptive but can require longer treatment periods and are strongly influenced by environmental conditions. Life-cycle and cost assessments are therefore preferable to simple labels such as 'low cost' or 'eco-friendly' (Table 2).

Table 2. Comparative Analysis of Phytoremediation with Conventional Remediation Technologies

Technology	Strength	Major limitation	Best suited to	Relative time
Phytoremediation	Low site disturbance; ecological restoration potential	Slow; site-specific; biomass management	Large, moderately contaminated sites	Long
Bioremediation	Effective for many biodegradable organics	Microbial activity is condition-dependent	Biodegradable organic pollutants	Moderate–long
Soil washing	Rapid physical contaminant removal	Secondary waste; high water/chemical demand	Selected contaminated soils	Short–moderate
Thermal treatment	Rapid destruction/removal of many organics	High energy demand and cost	Highly contaminated soils/materials	Short
Electrokinetic remediation	Useful for some metals and fine-grained soils	Energy demand and soil-specific limitations	Fine-grained contaminated soils	Moderate

6. Challenges and Limitations

6.1. Biological and Technical Constraints

The principal technical limitation is the dependence of remediation on plant establishment and growth. High contaminant concentrations may inhibit germination, photosynthesis, nutrient acquisition, and biomass production. Rooting depth can also restrict treatment to shallow contamination. Climate, drought, flooding, salinity, soil texture, nutrient status, and seasonal variation further affect performance. These constraints explain why strong laboratory results do not necessarily predict field-scale success^{7,8}.

6.2. Contaminant Bioavailability and Remediation Endpoints

Total contaminant concentration alone is an inadequate measure of phytoremediation potential. Plants respond primarily to the chemically and biologically available fraction, which varies with pH, organic matter, mineral surfaces, redox conditions, and competing ions. Remediation objectives must therefore distinguish between extraction, stabilization, degradation, and risk reduction. A site may be

successfully stabilized without achieving a large reduction in total contaminant concentration.

6.3. Contaminated Biomass Management

Harvesting contaminant-rich biomass creates a secondary waste-management problem. Uncontrolled decomposition, transport, composting, or inappropriate disposal may return contaminants to the environment. Potential post-harvest options include controlled thermal conversion, pyrolysis, gasification, secure disposal, and, where economically and technically feasible, metal recovery. Biomass management should therefore be incorporated into the original remediation design rather than treated as an afterthought^{15,16}.

6.4. Ecological and Regulatory Considerations

The introduction of non-native or genetically modified plants can create ecological concerns, including invasiveness, gene flow, food-chain transfer, and unintended effects on native communities. Phytovolatilization also requires explicit assessment of atmospheric exposure. Regulatory frameworks should define acceptable endpoints, monitoring requirements, biomass

disposal procedures, and long-term risk management for phytoremediation projects¹⁷.

6.5. Economic and Social Constraints

Although phytoremediation can reduce excavation and infrastructure requirements, it is not automatically inexpensive. Costs include site characterization, planting, irrigation, amendments, maintenance, harvesting, biomass transport, monitoring, and post-treatment management. The long treatment period may also conflict with stakeholder expectations for rapid remediation. Economic assessments should therefore include the complete treatment life cycle.

7. Emerging Innovations and Future Directions

7.1 Microbe-assisted phytoremediation

Selection of plant-growth-promoting bacteria, fungi, and mycorrhizal associations may improve plant establishment, contaminant availability, stress tolerance, and degradation.

7.2 Soil amendments and hybrid systems

Biochar, organic amendments, chelators, and mineral amendments can modify contaminant availability and improve plant growth. Their use should be evaluated for persistence, secondary pollution, and long-term stability.

7.3 Genetic and molecular improvement

Genome-wide approaches, functional genomics, gene editing, and marker-assisted selection may identify traits associated with uptake, translocation, tolerance, and biomass production. Field validation and ecological risk assessment remain essential.

7.4 Nanotechnology

Engineered and naturally derived nanoparticles are being investigated to alter contaminant availability and plant responses. Their own environmental fate and toxicity must be assessed before large-scale application.

7.5 Modelling and decision support

Mechanistic and process-based models can integrate soil chemistry, plant growth, contaminant transport, and environmental variability. Better integration of physical, chemical, and biological processes is needed for reliable prediction¹⁸.

7.6 Biomass valorization

Phytoremediation can potentially be linked to phytomining, energy recovery, or material recovery where contaminant and biomass characteristics make such pathways safe and economically viable.

8. Research Gaps and Priority Questions

- Which combinations of plant genotype, microbial consortium, and amendment provide reproducible field-scale performance?
- How can contaminant bioavailability and plant uptake be predicted under changing soil and climatic conditions?
- What standardized indicators should be used to compare phytoremediation performance across sites?
- How can contaminated biomass be managed with minimum secondary environmental risk and maximum resource recovery?
- When should phytoremediation be used alone, and when should it be integrated with physicochemical or microbial technologies?
- What long-term ecological and economic evidence is required before genetically improved plants or nanomaterials are deployed at scale?

9. Conclusion

Phytoremediation has evolved from a primarily plant-based contaminant-removal concept into an interdisciplinary field integrating plant physiology, soil science, microbiology, molecular biology, biotechnology, environmental engineering, and ecological restoration. Its principal strengths are low site disturbance, compatibility with ecological restoration, and the potential for cost-effective management of large areas with suitable contamination levels. Nevertheless, phytoremediation is inherently site-specific and often slow, and its effectiveness is constrained by contaminant bioavailability, plant growth, rooting depth, environmental conditions, and post-harvest biomass management. It should therefore be viewed as one component of a broader remediation toolbox rather than a universal replacement for conventional technologies. Future progress will depend on rigorous field validation, integrated plant–microbe and amendment strategies,

molecular and omics-assisted improvement, predictive modelling, life-cycle assessment, safe biomass management, and clear regulatory frameworks. A site-specific, risk-based, and integrated approach offers the strongest pathway for translating phytoremediation from experimental promise to reliable environmental practice.

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Author Contributions

AY: conceptualization, literature review, data curation, synthesis, figures and tables, and original drafting. **Md. MR:** literature review, data collection, validation, and critical revision. **AIA:** literature review and data collection. **SC:** conceptualization, supervision, methodological guidance, and critical revision. **AfA:** scientific guidance, validation, supervision, and critical revision. All authors approved the final manuscript.

Data Availability

No new experimental datasets were generated for this review. The synthesis is based on information available in the cited literature.

Use of Artificial Intelligence

Artificial intelligence tools were used to assist with language editing and manuscript organization. The authors reviewed and verified the scientific content and remain responsible for the final manuscript.

Conflicts of Interest

Authors state no conflict of interest.

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